Cerebellar-Inspired Neural Architectures for Motor Learning in Simulated Environments

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Outline

1. Motivation
2. Introduction
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Motivation

The Challenge

Learning motor skills requires coordinating movements over time, adapting to different goals and correcting errors based on feedback.

Supervised Learning

Uses prediction errors to improve movement execution.

Reinforcement Learning

Uses reward feedback to improve action selection.

Research Question

Can biologically inspired learning mechanisms improve the efficiency, reliability and generalization of motor learning?

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Introduction

This thesis will be based on two papers:

- Cerebellar-driven cortical dynamics can enable task acquisition, switching and consolidation $\rightarrow$ SL approach
- Policy-Gradient Reinforcement Learning as a General Theory of Practice-Based Motor Skill Learning $\rightarrow$ RL approach

Cerebellar-driven Cortical Dynamics Approach

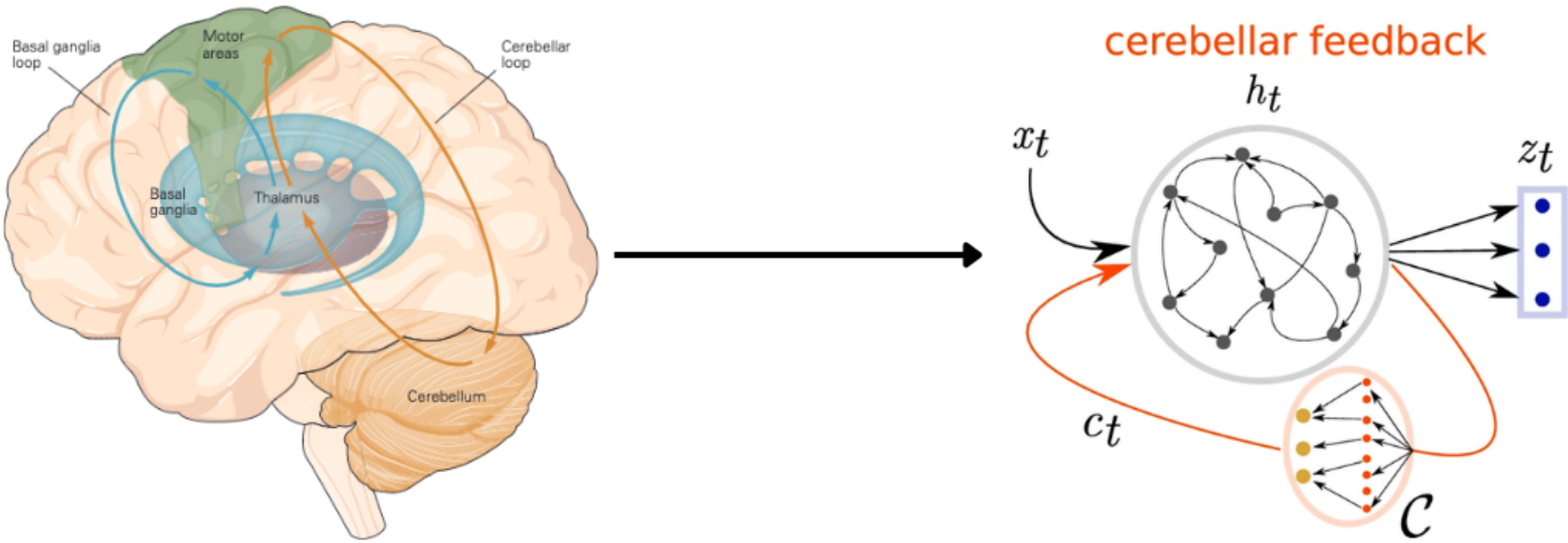
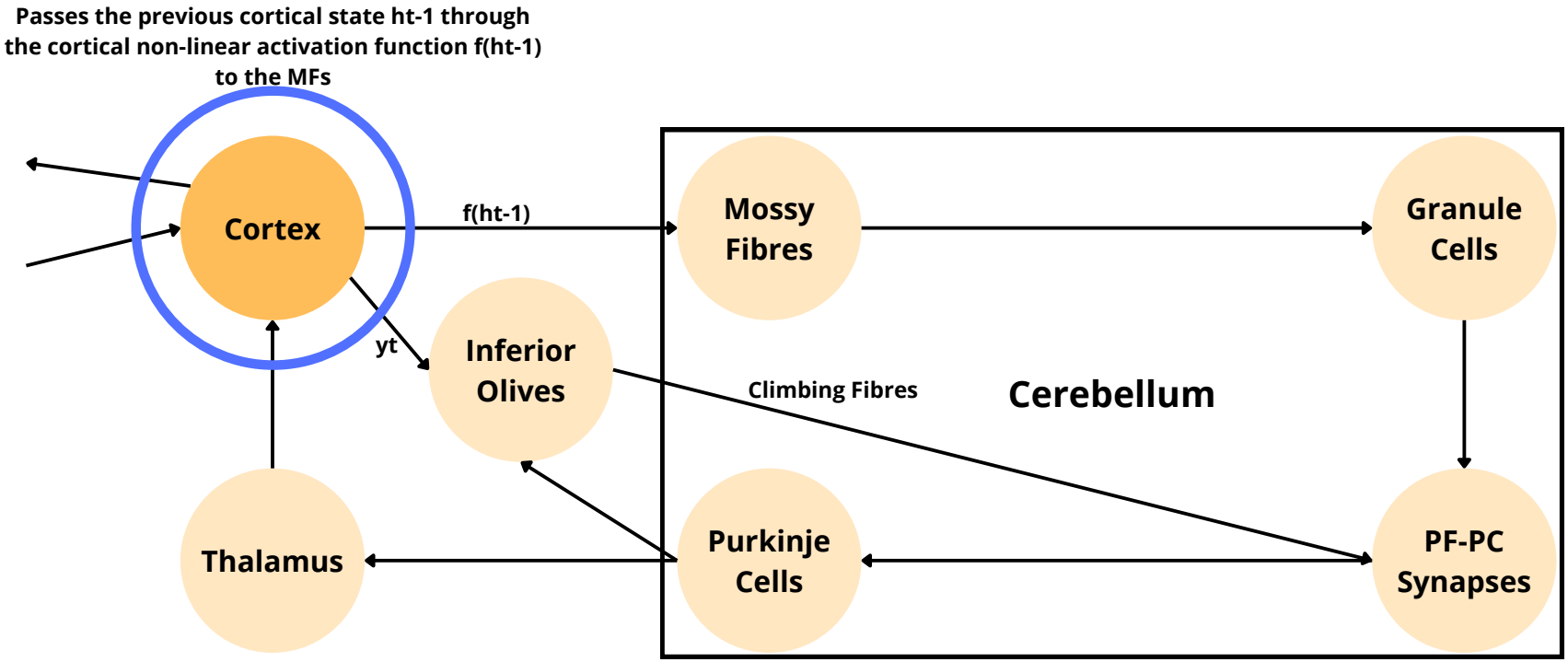
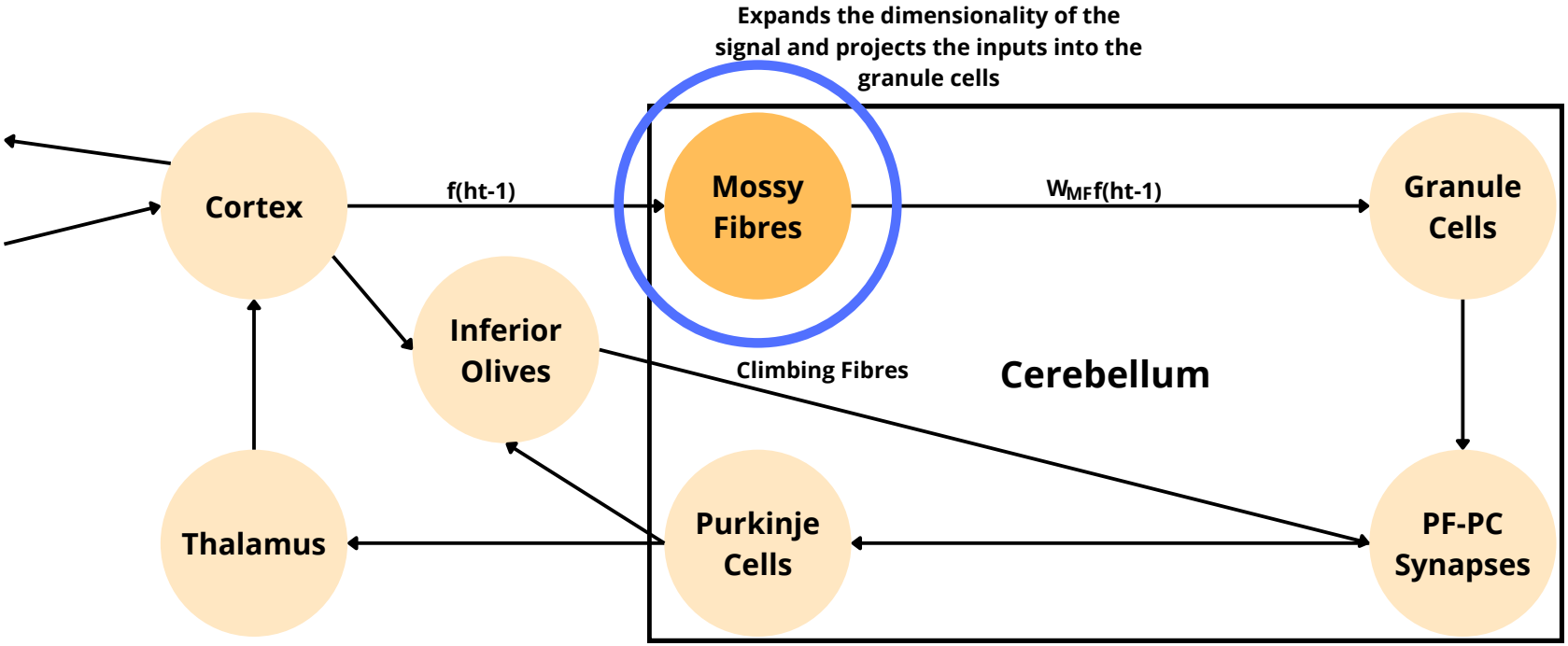


Figure: Cerebro-cerebellar network architecture

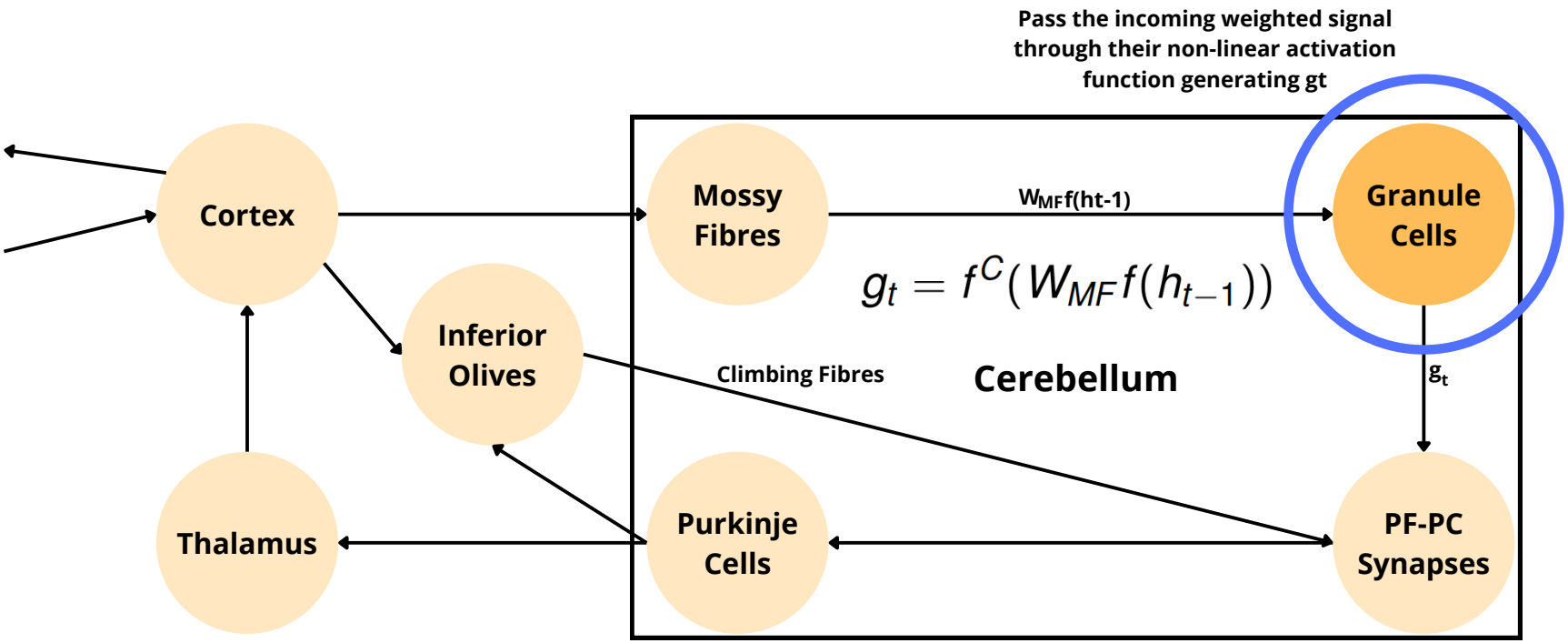
Modeled Cerebellar Feedback Loop



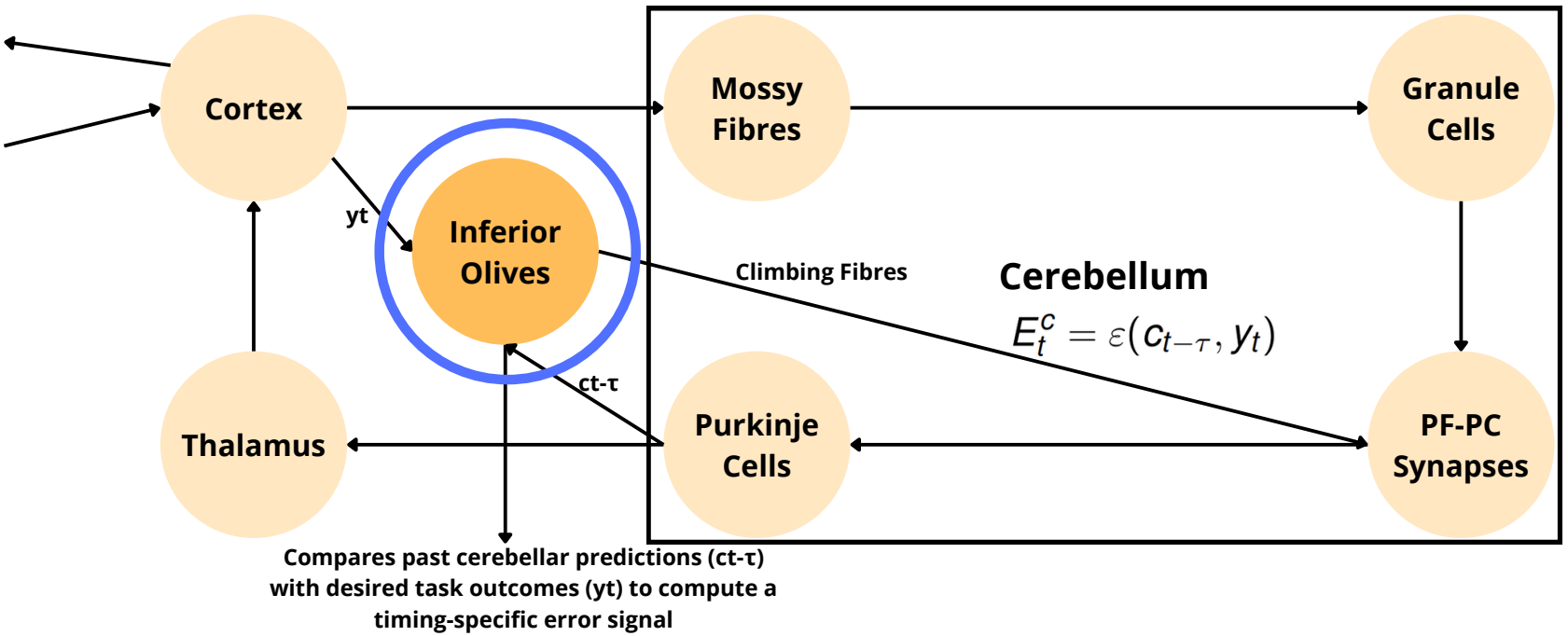
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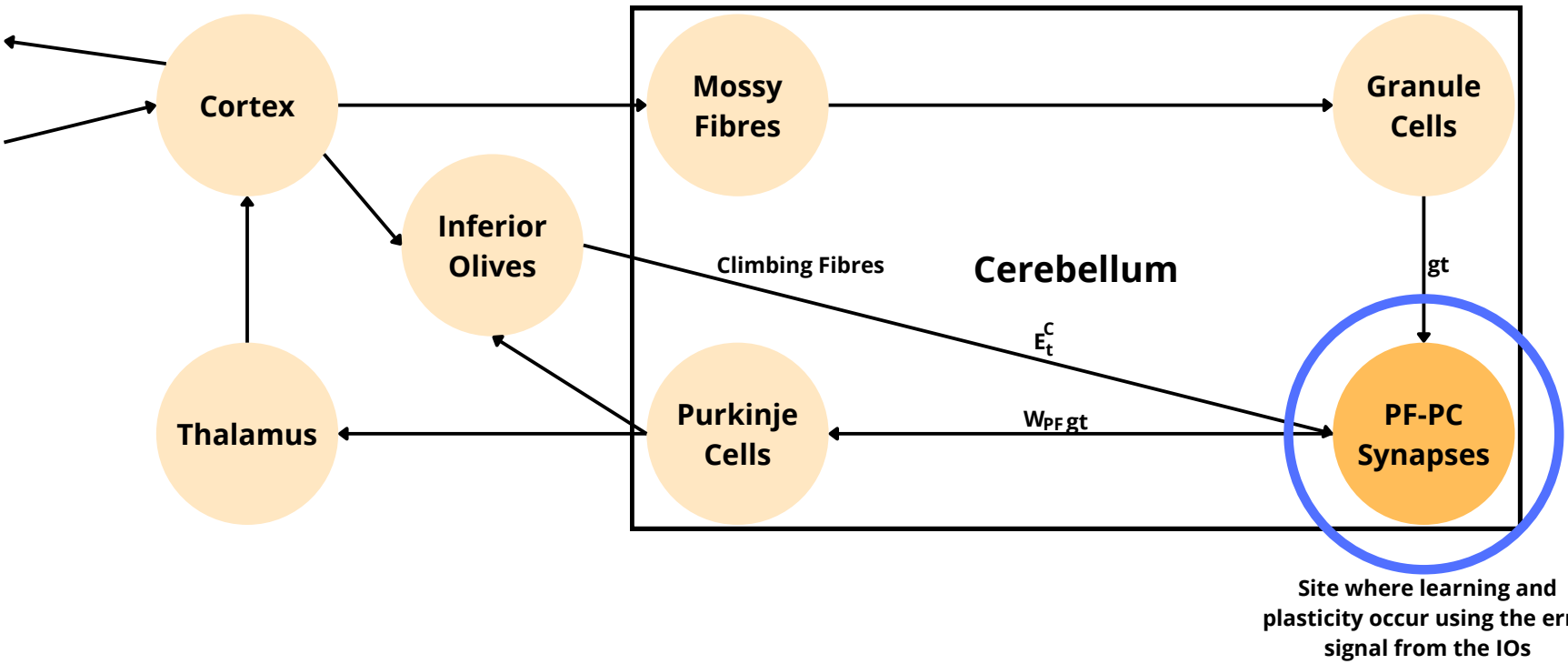
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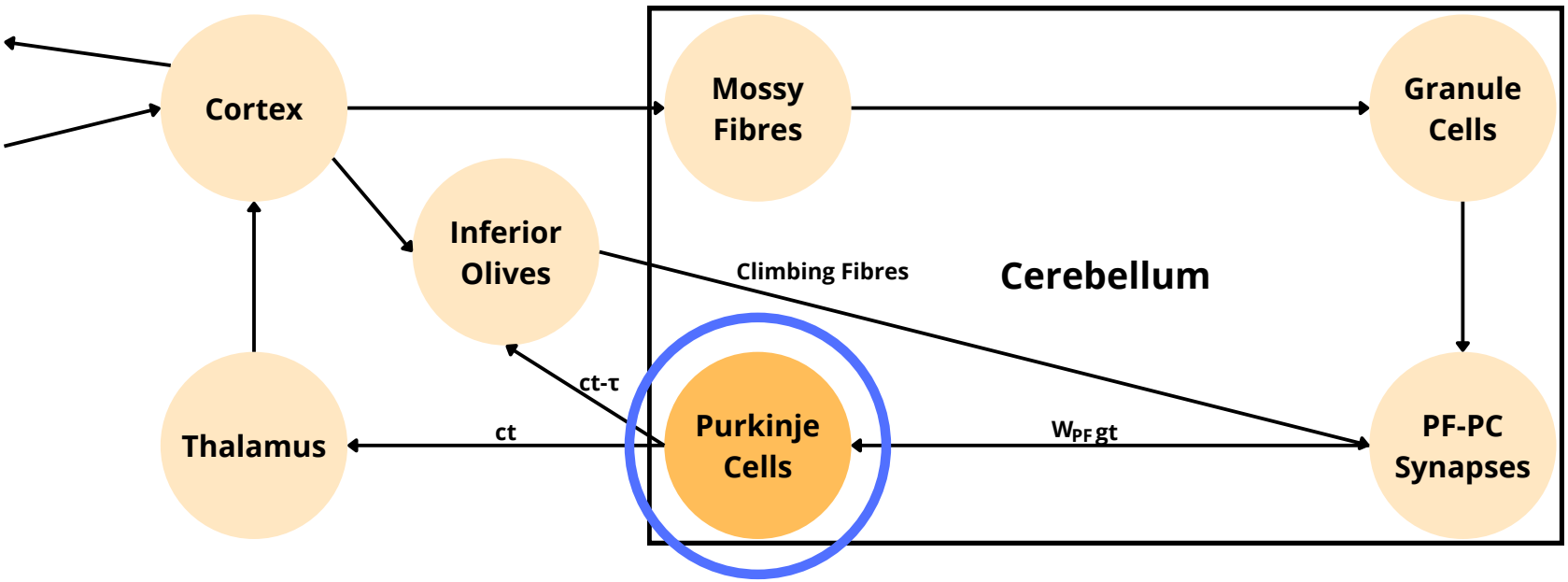
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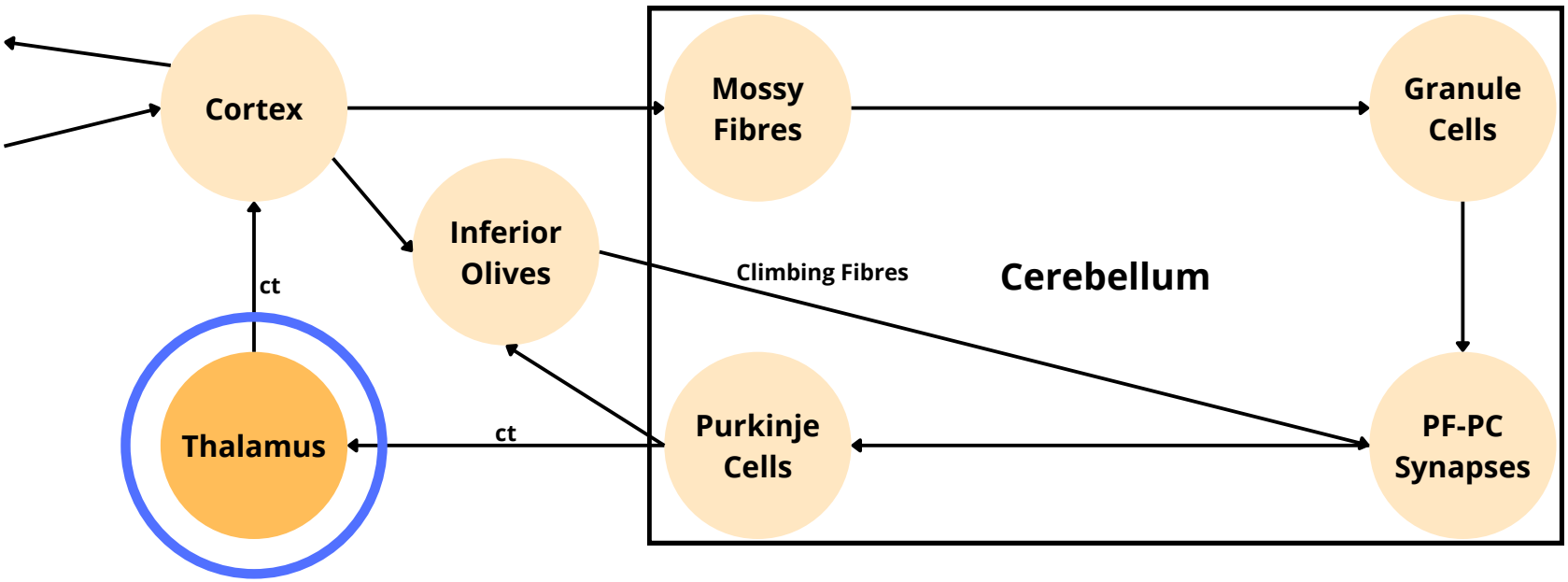
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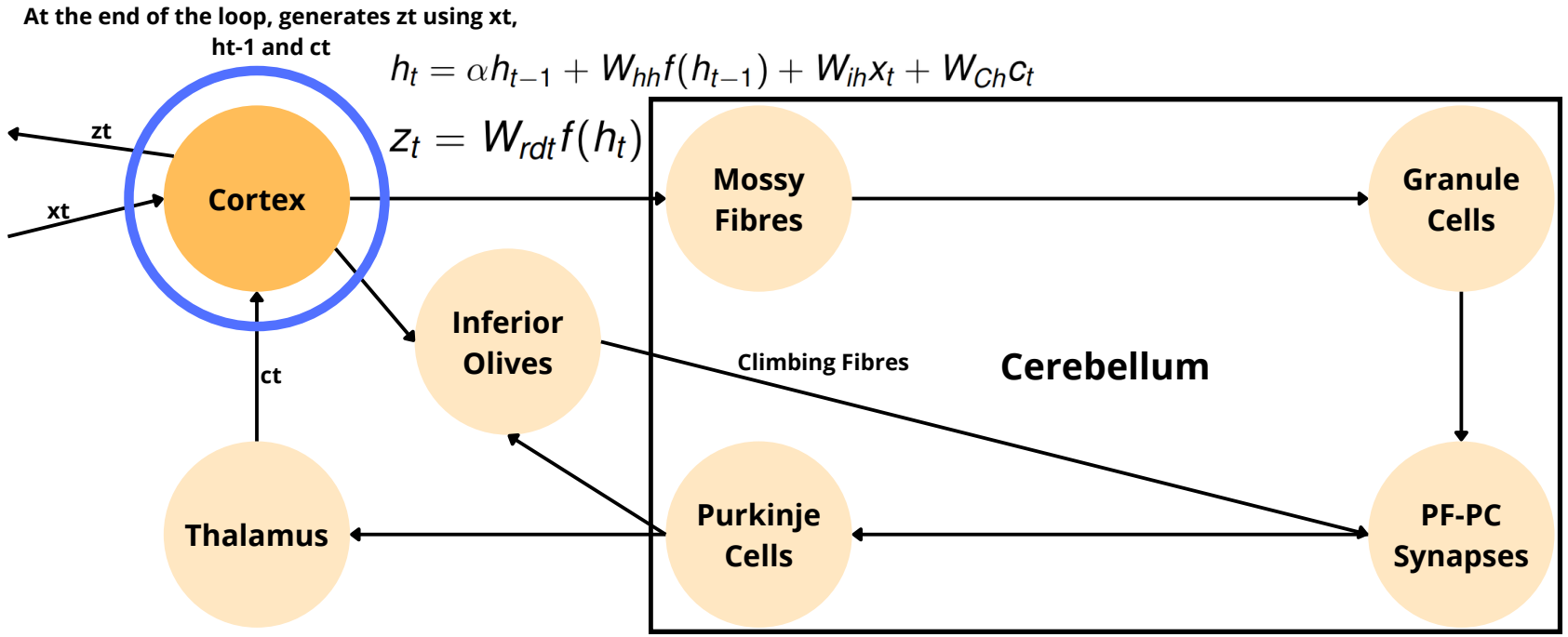
Integrates the signals from the synapses to generate task-specific predictions (ct ≈ yt+τ)

$$c_t = C(f(h_{t-1})) = W_{PF}g_t$$

Modeled Cerebellar Feedback Loop



Modeled Cerebellar Feedback Loop



Model Training Error Functions

Two types of error functions are used:

- **Task Error:**

$$E_t = \varepsilon(z_t, y_t) \tag{1}$$

- **Cerebellar Error:**

$$E_t^c = \varepsilon(c_{t-\tau}, y_t) \tag{2}$$

Cerebellar Feedback Model

Three types of cerebellar feedback models are considered:

- **Fixed RNN:** W_{hh}, W_{ih}, W_{MF} and W_{Ch} are fixed, while W_{rdt} and W_{PF} are trained.
- **Input Plastic:** W_{hh} and W_{MF} are fixed, while W_{ih}, W_{rdt}, W_{PF} and W_{Ch} are trained.
- **Fully Plastic:** All weights are trained.

Policy-Gradient RL Approach

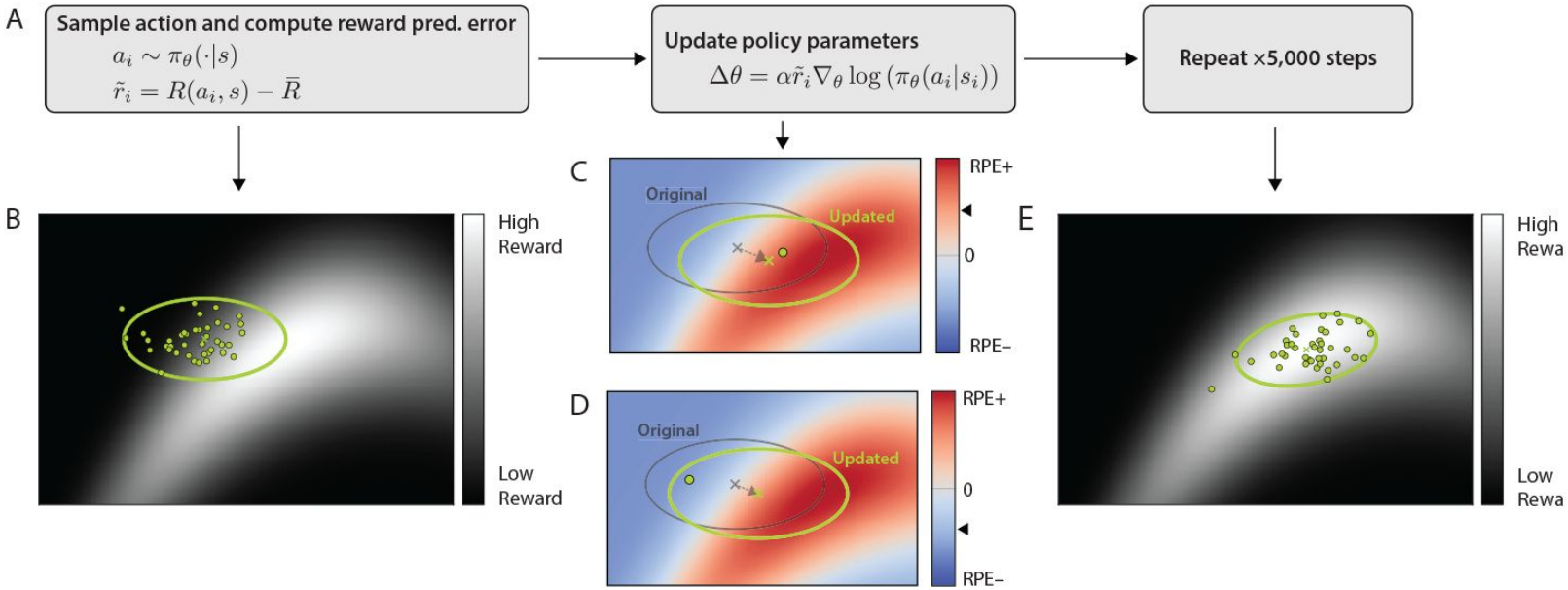


Figure: Illustration of policy-gradient RL

Model Computational Representation

- The model represented is of trial-by-trial skill learning:

$$\theta_{i+1} = \theta_i + \alpha(r(s_i, a_i) - \bar{r}_i)\frac{\partial}{\partial\theta}\log\pi(a_i|s_i). \tag{3}$$

- The average reward $\bar{r}_i$ is also computed on a trial-by-trial basis, according to

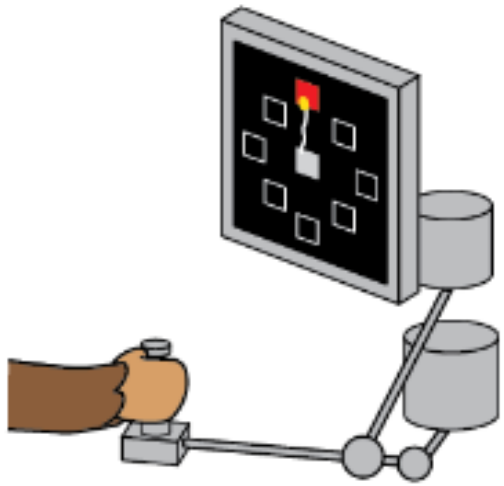
$$\bar{r}_{i+1} = (1-\beta)\bar{r}_i + \beta r(s_i, a_i). \tag{4}$$

- The policy $\pi(a_i|s_i)$ is assumed to be a Gaussian distribution, with mean μ and a covariance matrix Σ:

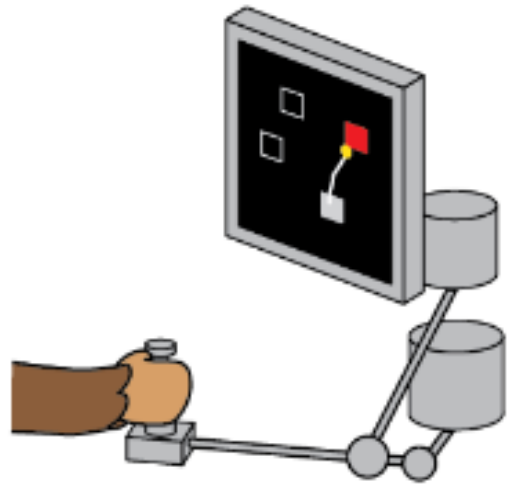
$$\pi(a|s) = \frac{1}{(2\pi)^{\frac{N}{2}}|\Sigma|^{\frac{1}{2}}}e^{-\frac{1}{2}(\mu-a)^T\Sigma^{-1}(\mu-a)}. \tag{5}$$

where N is the dimensionality of the action space.

Evaluation

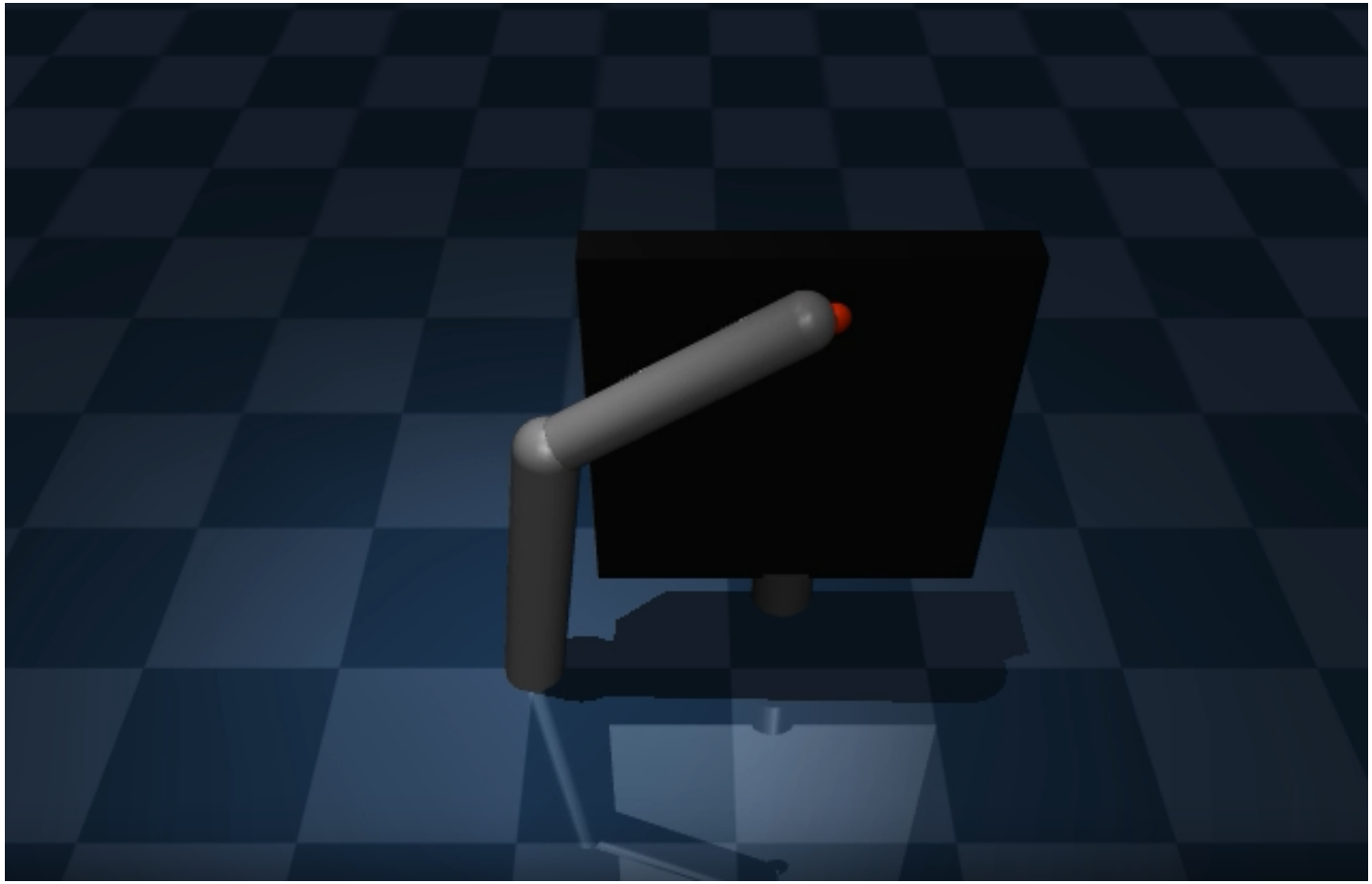


Center-out reach



Random target

Environment



Conclusion

After evaluating both of the paper's approaches, the end goal is to combine the SL and RL frameworks to create a hybrid architecture:

